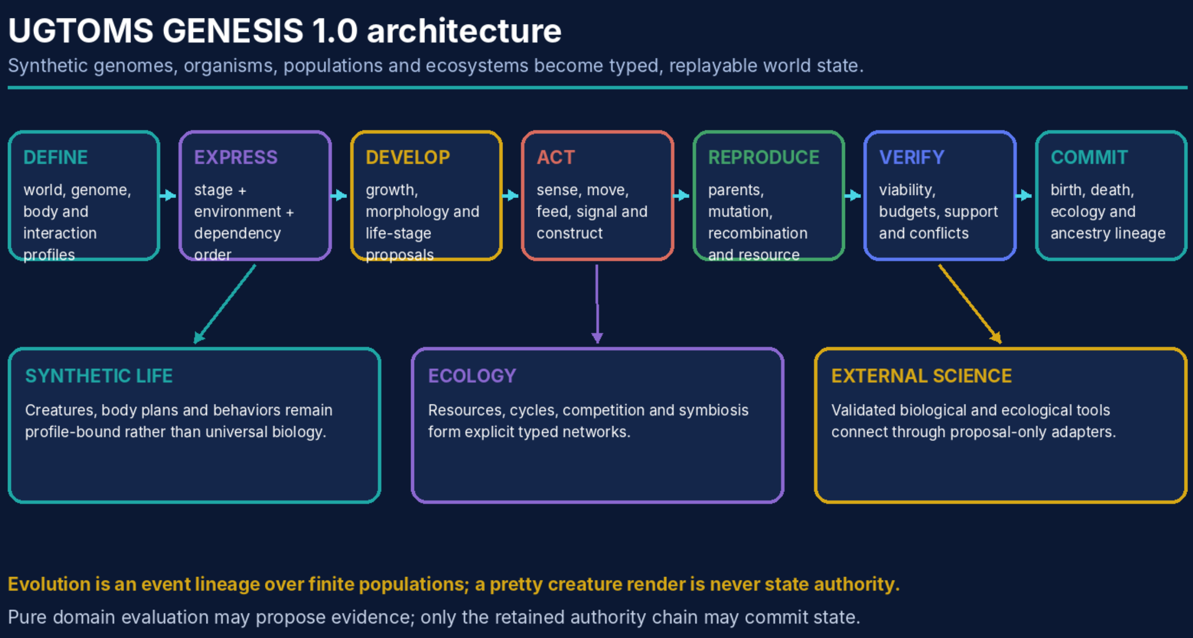
UNIVERSAL GEOMETRIC TOPOLOGICAL
OPEN MODULAR SUBSTRATE

UGTOMS GENESIS 1.0

Synthetic Life, Evolution, Morphogenesis and Ecosystem Cartridge



`ugtoms.domain.genesis@1.0.0`

Component	GENESIS
Canonical identity	ugtoms.domain.genesis@1.0.0
Component type	Domain cartridge / artificial-life world model
Version / date	1.0.0 / 30 August 2026
Document type	Open modular cartridge specification and implementation-ready profile
Status	Formal synthetic-life cartridge and deterministic fixtures; not a universal biological theory, laboratory protocol, medical genetics system or environmental certification

ATTRIBUTION AND EVIDENCE BOUNDARY

Prepared for Tom Klootwijk. Requester attribution is recorded as supplied and is not independently verified. This is a technical design artifact, not legal proof, scientific validation, regulatory approval or production safety certification.

Contents and Reading Rule

Section	Purpose
1	Executive decision and release scope
2	Source basis and evidence discipline
3	Open modular cartridge ABI
4	Formal synthetic-life state and authority classes
5	Genomes, traits and development programs
6	Bodies, morphology and life stages
7	Energy, matter, sensors and actions
8	Reproduction, mutation and population events
9	Selection, niches and ecological networks
10	Deterministic Glassreef scenario
11	Storage, queries, adapters and publication
12	Validation, boundaries and kill criteria
13	Mechanism catalog
14	Claims ledger
15	High-impact profiles and final definition

READING RULE

The UGTOMS source line supplies the reusable authority, typing, order, evidence and packing discipline. The specialist domain model in this cartridge is an engineering-derived extension. Where the source corpus is silent, this report says so instead of presenting general domain knowledge as source-derived.

Executive Decision and Release Scope

FORMAL DECISION

GO for GENESIS as a synthetic-life and ecosystem cartridge. The decisive move is to treat genomes, development programs, organisms, populations, resources and ecological interactions as typed definitions and verified events rather than a pile of decorative creature parameters. Birth, mutation, growth, feeding, death and ancestry become replayable world changes with explicit budgets and reason codes.

GENESIS is deliberately about artificial and educational life worlds first. It can host evolving games, creature construction, procedural ecologies and research-shaped simulations, but it does not pretend that one compact profile captures real molecular biology, population genetics or ecology. Mature scientific models stay behind adapters with their own units, assumptions, calibration and validation.

- A genome is a stable generative definition; an organism is a lineage-bearing individual state. They are never collapsed into one identity.
- Phenotype is produced by genome, stage, development order, environment and declared stochastic inputs. A single gene-to-trait shortcut is never assumed globally.
- Mutation and reproduction are proposals. Failed viability or resource guards create reason-coded candidate evidence without creating ghost offspring.

- Evolution is the accumulated consequence of births, deaths, interactions, inheritance and branch history under one explicit world profile.

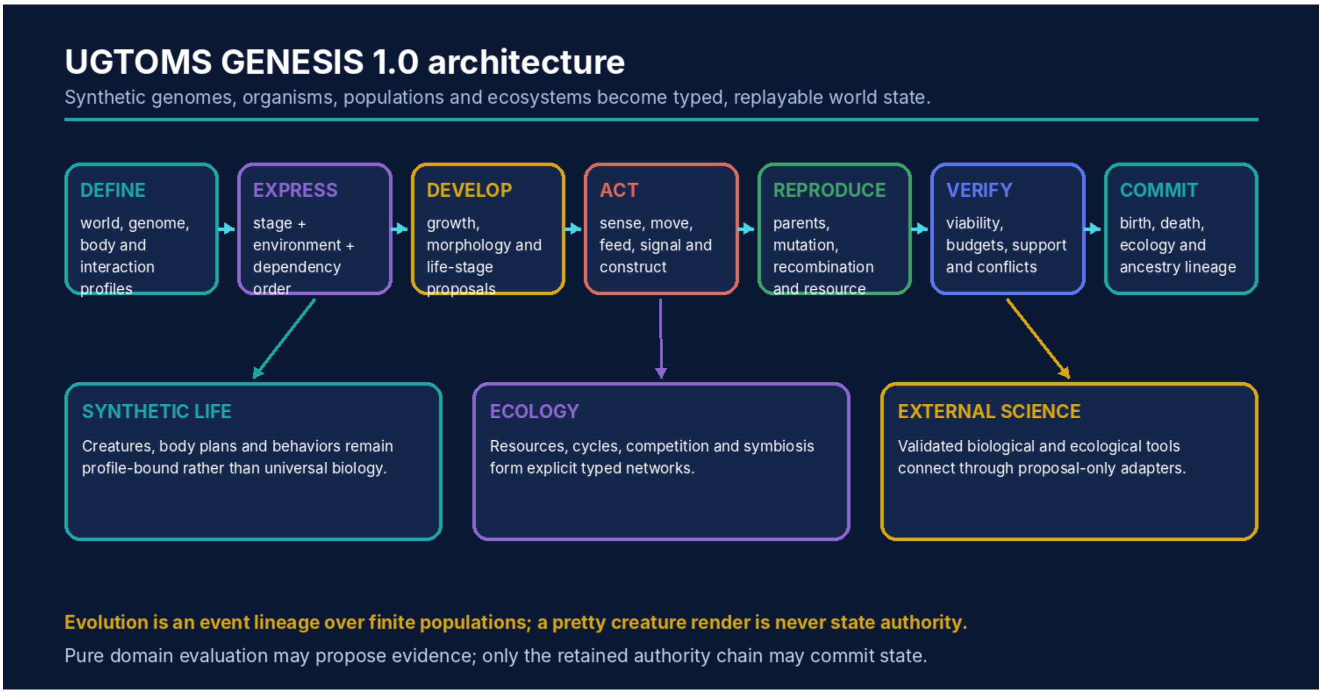


Figure 1. GENESIS maps artificial-life domain operations into the retained UGTOMS authority chain.

Source Basis and Evidence Discipline

The cartridge inherits the following UGTOMS commitments:

- Typed definitions and operators live inside the substrate, with stable IDs, dependencies and content hashes [S1-S3].
- Pure evaluation is upstream of support, compatibility, guard, verified proposal, deterministic commit and lineage [S1, S2, S4].
- Coordinates, pictures, sounds, meshes and packed codes are not identity or authority by themselves [S4-S8].
- Uncertainty, exactness, capability and error margins are explicit and may only be strengthened by evidence [S1, S2, S8].
- Component-scoped identities prevent a specialist domain pack from masquerading as a universal replacement of every other release [S10].

ENGINEERING-DERIVED DOMAIN CONTENT

The genome, development, metabolism, organism, population, reproduction, mutation, selection and ecosystem records in GENESIS are engineering-derived. The registered UGTOMS sources contribute typed definitions, ordered evaluation, finite state, event authority, evidence boundaries, clocks, packing, replay and external-adapter discipline [S1-S15]. REACT contributes a useful separation between symbolic definitions and observed batches, but GENESIS does not claim that chemistry alone determines a complete living system [S10].

ID	Registered source	Use in this cartridge	Pages
S1	UGTS_FOUNDATION_UNICODE_v5.pdf	Foundation 5.0 OperatorCells: literal, glyph, semantic, field, converse, packing and content-addressed faces remain typed.	pp. 1-21
S2	UGTS_KC_4_2_General_Operator_Order_Addendum(1).pdf	Typed OperatorSpec, dependency/phase/event/replay order, exactness, effects, capabilities, budgets and adapter boundaries.	pp. 2-32
S3	UGTS_KC_3_6_Tom_Klootwijk.pdf	Definitions as content-addressed substrate data, resolvable references, dependency closure and explicit operation order.	pp. 1-15
S4	Unified_Geometric_Topological_Substrate_GPU_Native_Addendum.pdf	Canonical support → compatibility → guard → verified event → transition → lineage authority and packed-state limits.	pp. 2-15
S5	UGTS_KC_2_0_Tom_Klootwijk_Expanded_Substrate.pdf	Patterns, fields, topology, SE(2)/SE(3), bounded dynamics, hybrid events and numerical/error contracts.	pp. 3-15
S6	UGTS_KC_Two_Hands_3_0_Interactive_Graphics_Runtime.pdf	Stable scene identity, proposals, deterministic commit, checkpoints, rollback, replay and divergence detection.	pp. 3-19
S7	UGTS_KC_3_9_KC_Elizabeth_Vector_Game_Runtime.pdf	Versioned projects, deterministic fixed-step runtime, snapshots/hashes and self-contained browser delivery.	pp. 2-16
S8	UGTS_KC_3_6_2_SCLP_Tom_Klootwijk.pdf	Time/phase/winding, finite topology maps, interval guards, finite grammar and exact packed-key boundaries.	pp. 1-15
S9	UGTS_Versioning_Charter_Phase_1_Foundation_Chronology(1).pdf	Component-scoped identities and explicit release graphs instead of one misleading global ladder.	pp. 1-6
S10	01_UGTOMS_REACT_1_0.pdf	Typed species/composition-like records, conservation checks, batch/evidence separation and external scientific adapters.	pp. 2-15
S11	04_UGTOMS_RULEFORGE_1_0.pdf	Reusable rule-world host: typed actions, visibility, chance, conflict resolution, terminality and deterministic replay.	pp. 2-18
S12	06_UGTOMS_PROOFWEAVE_1_0.pdf	Proof graphs, small kernel, scoped certificates, trust boundaries and proposal-only search.	pp. 2-18
S13	07_UGTOMS_CIVITAS_1_0.pdf	Layered world identity, route/support graphs, operational dependencies, observations, scenarios and verified city patches.	pp. 2-19
S14	08_UGTOMS_ARCANA_1_0.pdf	Typed fictional world laws, resource ledgers, executable symbols, support fields and rule-composition discipline.	pp. 2-20
S15	09_UGTOMS_CHRONOFORGE_1_0.pdf	Multiple clocks, immutable events, branch forks, loops, compensation, descendant invalidation and counterfactual replay.	pp. 2-20

Open Modular Cartridge ABI

The cartridge is a concrete instance of the shared domain object D_GENESIS:

```
D = (T, O, R, U, I, Q, E, G, Delta, L, Cert, K, Pi)
```

Field	Contract
T	Entity, value and state types
O	Typed operators and transformations
R	Relations, graphs, topology and converse pairs
U	Units, dimensions, coordinate and profile conventions
I	Invariants, conservation laws and constraints
Q	Dependency, precedence, phase and reduction order
E	Observations, evidence, uncertainty and provenance
G	Guards, thresholds and admission predicates
Delta	Proposed and committed state patches
L	Lineage, replay, history and migration
Cert	Exactness, residual, interval, scope and status certificates
K	Cold atlas, hot codebook and packed representation
Pi	Optional visual, audio, physical or device projection

A conforming cartridge contains a manifest, type registry, operator registry, relation atlas, unit/profile system, dependency DAG, guard library, transition definitions, claims ledger, examples and validation fixtures. External mature libraries may sit behind adapters when they are safer or more accurate than a fresh reimplementation.

General Operator Record and Order

```
OperatorSpec = (id, version, syntax, arity, domains, codomain, units, shape,
                laws, exactness, effects, capabilities, dependencies,
                order_contract, invariants, provenance, content_hash)
```

The record follows the General Operator and Order layer [S2]. A symbol or glyph is not allowed to smuggle in undeclared type, unit, order, inverse or authority semantics.

EXAMPLE RECORD

genesis.reproduce consumes parent organism states, reproduction profile, environment support, resource budgets and a deterministic stochastic context. It returns a rejected proposal or an offspring genome/body-state proposal plus a complete trace. It cannot directly allocate an organism ID, spend inventory or extend ancestry before viability and conflict guards pass.

Normative phase schedule

Phase	Meaning
1-3	parse, normalize and resolve content-addressed references
4-6	type, shape and unit validation; dependency closure
7-9	construct candidates; apply declared composition and reduction order
10-12	evaluate exact or bounded results; issue capability and residual certificate
13	support, compatibility and guard classification
14	verified proposal and deterministic conflict resolution
15	atomic commit, lineage, replay and optional projection

Formal Synthetic-Life State and Authority Classes

```
q_GNS = (P, G, O, B, E, N, F, S, H, L, X)

P world/biology/evolution profiles   G genome + development definitions
O organism and population states    B body/morphology/life-stage records
E energy, matter and inventory       N ecological relation network
F environment/resource/hazard fields S stochastic profile + seed/draw lineage
H checkpoints + current state hash   L births, deaths, mutations and ancestry
X observations, publication and external adapters
```

Authority class	May contain	Promotion boundary
World definitions	Genome modules, body plans, development rules, interactions, units and clocks.	Content-addressed definitions must resolve and validate before use.
Authoritative life state	Living/dead organisms, energy, body state, population membership, resources and ecology edges.	Only verified life/ecology events may change it.
Candidate variation	Mutation, offspring, behavior, niche and interaction proposals.	No stable offspring or resource spend before commit.
Observed evidence	Measured or user-supplied traits, environments, population counts and provenance.	Association, uncertainty and profile guards are required.
Scenario branch	Alternative seed, selection pressure, climate cycle or intervention.	A branch remains simulation until explicitly promoted.

IDENTITY SPLIT

genome_id != organism_id != body geometry != species label != population membership != screen sprite. A clone may share a genome while retaining a different birth event, history, state and fate.

Genomes, Traits and Development Programs

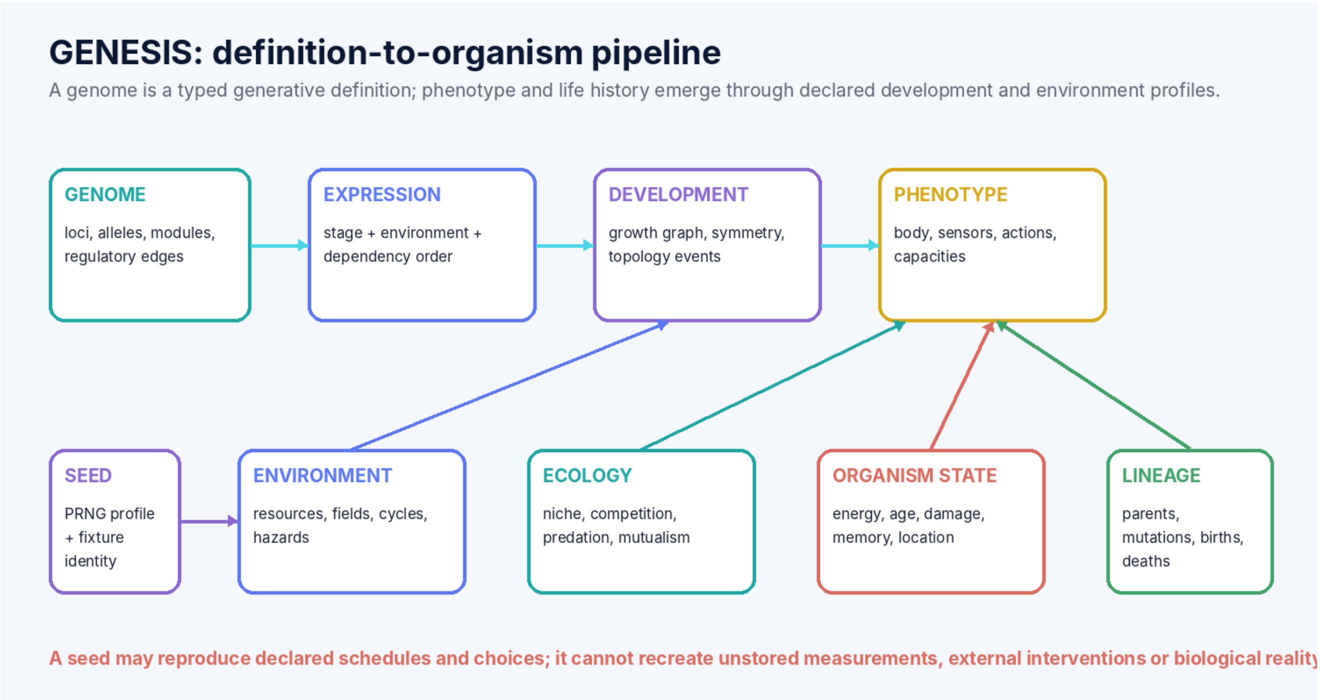


Figure 2. A genome resolves through expression and development profiles into phenotype and organism state; environment and lineage remain separate inputs.

```
Genome = (profile, loci, alleles, modules, regulatory_edges, invariants, provenance, hash)
ExpressionProposal = evaluate(genome, life_stage, environment, random_context)
DevelopmentProposal = ordered_modules(ExpressionProposal, resource_budget, body_state)
```

Record	Core contract	Boundary
Locus / allele	Typed choice or value with domain, alternatives, provenance and mutation contract.	No universal dominance, penetrance or molecular interpretation is inferred.
Regulatory edge	Dependency/condition controlling expression or module activation.	Cycles require an explicit fixed-point or recurrent profile.
Development module	Ordered growth, differentiation, symmetry or topology operator graph.	A module proposes state; it cannot bypass resource and viability guards.
Trait	Declared phenotype feature, measurement or capability.	A trait name is not an exact scientific ontology unless an external profile supplies it.
Expression trace	Which definitions, conditions, random draws and modules contributed to a phenotype proposal.	A trace explains the model, not biological truth outside the profile.

Bodies, Morphology and Life Stages

Layer	Representation	Examples of guarded change
Body graph	Finite labeled parts, joints, interfaces, tissues/materials and support relations.	Grow segment, add leaf, shed shell, heal connection.
Geometry profile	Parametric curves/fields/meshes or direct procedural projection tied to the body graph.	Scale limb, bend spine, unfold sail, thicken armor.
Symmetry/segmentation	Declared generative repetition with exceptions and identity mapping.	Bilateral mutation, radial bud, segment loss.
Life stage	Stage machine with entry/exit guards and available operators.	Hatch, metamorphose, mature, become dormant.
Topology event	Split, merge, budding, shedding or attachment with explicit identity migration.	Budding creates a child; shedding creates debris/resource, not a silent delete.

```
body_change verified only when support + stage + dependencies + resource budget
+ geometric/topological validity + conflict policy + viability profile pass
```

MORPHOLOGY IS NOT IDENTITY

A body can grow, lose parts, molt or metamorphose while the organism remains the same lineage node. Conversely, a visually identical clone is a different organism because its birth and event history differ.

Energy, Matter, Sensors and Actions

Subsystem	Typed state/operator	Required discipline
Energy/upkeep	Production, storage, basal cost, action cost, growth and reproduction reserves.	Quantities, units and external sources are explicit; no negative inventory.
Matter/nutrients	Resource classes, intake, conversion, waste and environmental return.	Selected conservation profile and process capability are explicit.
Sensors	Support query plus uncertainty/noise and organism-local observation.	A sensor reading is not direct world authority or guaranteed belief.
Actions	Move, feed, signal, defend, construct or manipulate proposals.	Body capabilities, cost, support, targets and conflicts must pass.
Memory/learning	Bounded organism-local event or feature records.	Memory cannot contain a fact that was never acquired through a declared operator.

```
action_ok = body_capability AND environmental_support AND target_compatibility
            AND energy/matter_budget AND timing_guard AND conflict_resolution
```

EXTERNAL MODELS

Detailed biomechanics, neural control, metabolism or chemistry may be supplied by specialist adapters. GENESIS preserves their capability and uncertainty labels instead of flattening every model into one fake universal creature equation.

Reproduction, Mutation and Population Events

Stage	Proposal content	Commit effect
Parent selection	Parents, roles, compatibility, stage, resources and authorization.	None; selection is still candidate context.
Genome construction	Clone, recombination, crossover, mutation draws and resulting content hash.	None; candidate genome remains unattached.
Development seed	Initial body/development state, environment and viability estimate.	None; no organism exists yet.
Guard sequence	Support, readiness, resource, capacity, viability and conflict status.	Pass, reject or unknown with reason trace.
Birth commit	Stable organism ID, parent edges, resource patches, population membership and event hashes.	Ancestry and world state extend atomically.

```
offspring_id = stable_id(world_seed, parent_ids, birth_event_index, genome_hash)
mutation draws are stored or reproducible from the declared PRNG schedule
rejected candidates do not consume authoritative resources or population capacity
```

STOCHASTIC HONESTY

A fixed pseudo-random profile is useful for deterministic games and fixtures. It is not a claim of physical randomness, biological mutation rates or universal evolutionary statistics.

Selection, Niches and Ecological Networks

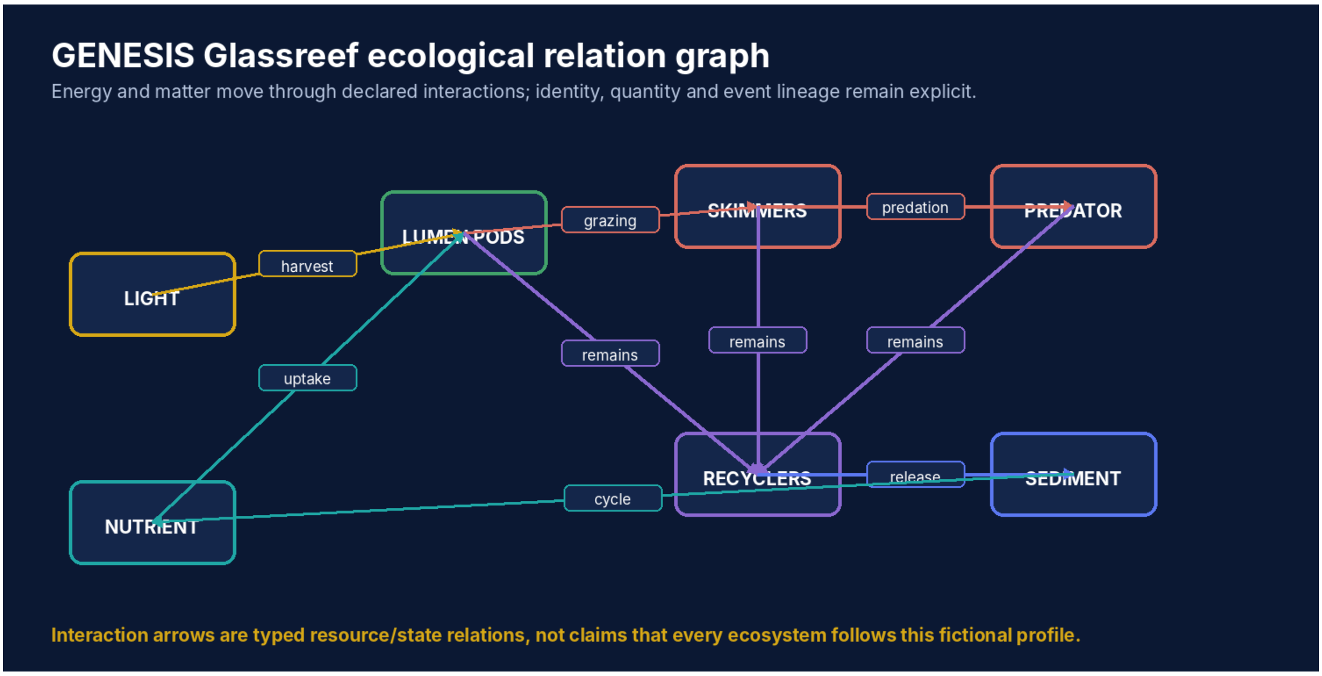


Figure 3. Glassreef uses explicit resource and interaction edges; the profile is fictional and inspectable.

Ecological concept	UGTOMS record	Important distinction
Niche	Habitat support + resource use + timing + interaction/capability profile.	A niche is not just a coordinate or species label.
Selection pressure	Profile-bound consequences affecting survival/reproduction under current environment.	It is contextual and can change over time.
Competition	Shared limiting resource plus encounter and allocation model.	Co-location alone does not imply competition.
Predation/grazing	Consumer-target relation with support, action, defense and resource transfer.	A relation does not guarantee every encounter succeeds.
Mutualism/symbiosis	Reciprocal exchange/benefit with persistence and dependency conditions.	Benefit may be asymmetric or conditional.
Drift/bottleneck	Population sampling and lineage effect under a declared stochastic/event profile.	No evolutionary improvement is implied.

Deterministic Glassreef Scenario

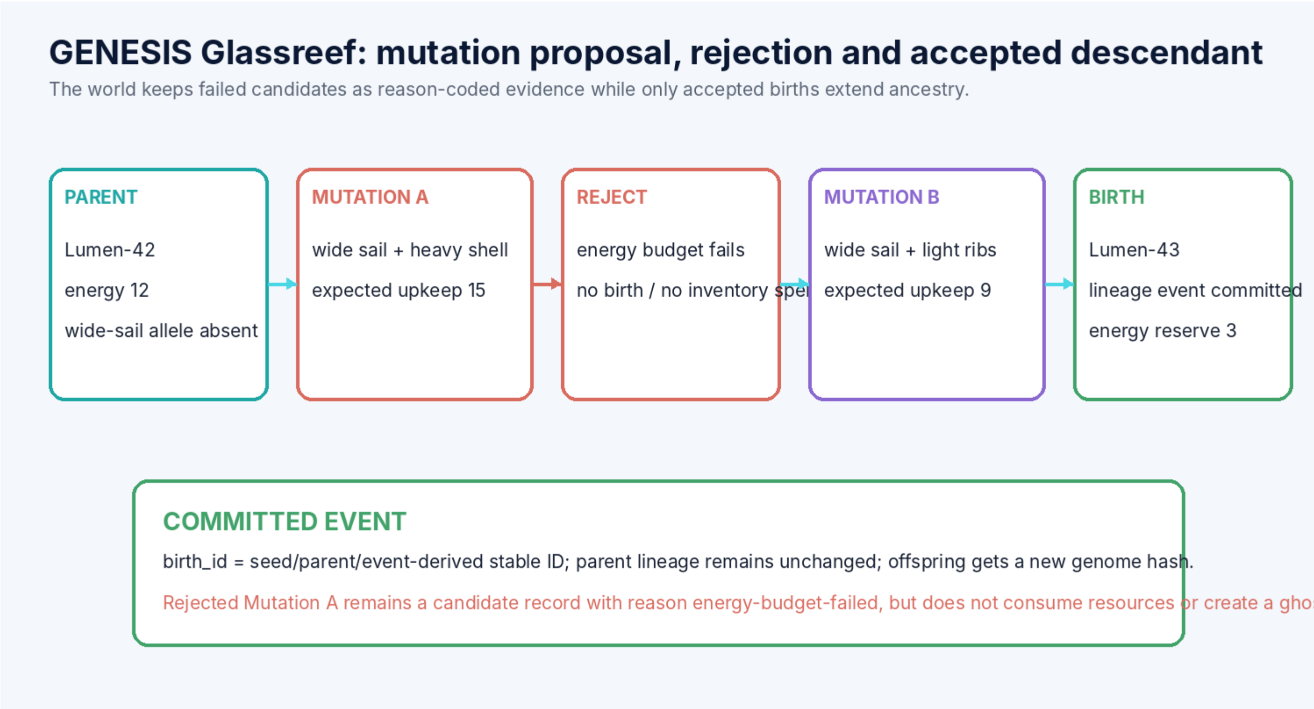


Figure 4. A costly mutation is rejected without a birth; a viable alternative creates one committed descendant and a new ancestry edge.

```
world profile: Glassreef-1, 12-tick light cycle, finite nutrient pool
parent: Lumen-42, energy=12, stage=adult
Mutation A upkeep=15 -> REJECT energy-budget-failed
Mutation B upkeep=9, birth cost=9 -> PASS; offspring Lumen-43, reserve=3
post-state hash binds genome, organism, resource and ancestry patches
```

Check	Fixture result
Definition closure	World, genome, development, energy and mutation operators resolve by hash.
Candidate isolation	Mutation A remains a rejected candidate and creates no organism ID.
Resource accounting	Mutation B spends exactly the declared reproduction cost and keeps nonnegative reserve.
Ancestry integrity	Lumen-43 has one birth event and a parent-of edge from Lumen-42.
Replay	Fixed seed, event order and profile regenerate the same accepted/rejected sequence.

Storage, Queries, Adapters and Publication

Interface	Contract
Genome/body cold atlas	Content-addressed definitions, modules, traits and geometry profiles; hot organisms store compact references.
Population snapshots + deltas	Periodic state snapshots plus birth/death/mutation/action/ecology events and external novelty.
Ancestry query	Parents, descendants, common ancestor, mutation provenance, trait derivation and first divergence.
Ecology query	Resource flow, interaction neighborhood, niche support, population change and unresolved model state.
RULEFORGE adapter	Turns organisms and ecosystems into game entities/actions without changing life-state authority semantics.
REACT adapter	Optional chemistry/stoichiometry layer for explicit resource transformations, never a hidden universal metabolism.
CIVITAS adapter	Places synthetic ecosystems inside layered habitats, infrastructure or changing environments.
External science adapter	Validated genetics, ecology, biomechanics or simulation software proposes typed results with version/provenance.

PACKING BOUNDARY

Compact genome hashes, seeds and codebook slots identify declared definitions and deterministic choices. They do not compress every possible phenotype, environment, observation or future lineage into one number.

Reference Validation, Boundaries and Kill Criteria

REFERENCE SELF-CHECK RESULT

38/38 deterministic specification self-checks passed. These checks cover catalog uniqueness, required fields, dependency closure, demo arithmetic and declared reason codes. They are package evidence, not independent domain validation.

Checked area	Result
Catalog integrity	48 unique GNS IDs with complete component-local contracts.
Genome/organism split	Two organisms may share a genome while retaining distinct birth and state identities.
Candidate isolation	Rejected mutation produces no resource spend, organism, population membership or lineage patch.
Birth atomicity	Accepted offspring creates genome reference, organism state, parent edge and resource delta together.
Seed replay	Fixed PRNG/profile/event order regenerates the Glassreef fixture.
Budget guard	Growth, action and reproduction cannot commit negative energy or matter inventory.
Ecology typing	Predation, competition and mutualism remain distinct relations and do not arise from proximity alone.

Explicit non-claims

- No universal molecular, genetic, developmental, ecological or evolutionary theory is claimed.
- No medical, reproductive, conservation-management, biosecurity or laboratory guidance is supplied.
- No stochastic profile is claimed to reproduce physical randomness or measured mutation rates.
- No generated phenotype image, score or fitness value proves real biological capability.
- No seed reconstructs unstored observations, interventions, environmental novelty or human choices.

Kill criteria

1. Reject or narrow a use when a symbol, codebook slot or packed record cannot resolve to its exact global definition.
2. Reject when a capability or exactness label is stronger than the model, measurement or external source supports.
3. Reject when units, profiles, coordinate systems, operand order or dependency order are implicit.
4. Reject when a proposal bypasses support, compatibility, guards, deterministic commit or lineage.
5. Reject when an external adapter is required for safety or conformance but its version, calibration or provenance is absent.
6. Reject any performance, compression or safety claim without a named workload, target, error/equivalence contract and evidence.

Complete GENESIS 1.0 Mechanism Catalog

The catalog contains 48 cartridge-local mechanisms. These IDs are component-scoped and do not silently extend a global M-number ladder.

ID	Family	Mechanism	Core contract
GNS001	Profile	Synthetic-life world profile	Declares organism, development, environment, stochastic, quantity and event semantics.
GNS002	Identity	Genome identity	Stable content-addressed generative definition; display string and organism identity remain distinct.
GNS003	Identity	Organism identity	Stable life-history node with birth event, parent links and current state.
GNS004	Identity	Population identity	Typed grouping rule, membership profile and revision lineage.
GNS005	Identity	Species profile	Optional finite classification/compatibility profile; not a universal biological species law.
GNS006	Definition	Locus definition	Typed parameter or operator selector with domain, alternatives and provenance.
GNS007	Definition	Allele value	Profile-bound value at a locus; no implicit dominance or expression semantics.
GNS008	Definition	Regulatory edge	Dependency/condition relation controlling declared expression order.
GNS009	Definition	Development module	Content-addressed growth/morphogenesis operator graph with stage and resource contract.
GNS010	Definition	Trait definition	Typed phenotype feature with measurement, capability and uncertainty profile.
GNS011	Morphology	Body graph	Finite labeled parts/interfaces/topology; geometry is a projection under a declared body profile.
GNS012	Morphology	Symmetry profile	Radial, bilateral, segmented or custom generative symmetry with explicit exceptions.
GNS013	Morphology	Growth schedule	Stage/time/resource-dependent body transformation sequence.
GNS014	Morphology	Life-stage transition	Guarded transition between embryo, juvenile, adult, dormant or custom stages.
GNS015	Morphology	Topology change	Declared split, merge, budding, shedding or metamorphosis event with identity migration.
GNS016	State	Energy ledger	Finite resource quantity and budgets for upkeep, action, growth and reproduction.
GNS017	State	Matter inventory	Typed consumable/material quantities; no creation by unbalanced state patch.
GNS018	State	Damage and repair state	Profile-bound integrity variables, repair operators and irreversible markers.
GNS019	State	Sensor state	Observed signal estimates with support, noise/uncertainty and capability.
GNS020	State	Memory state	Organism-local retained event/feature records under a bounded memory profile.
GNS021	Environment	Resource field	Finite spatial/temporal resource support with quantity and regeneration profile.
GNS022	Environment	Hazard field	Typed cost/damage/guard relation; presentation intensity is not authority.
GNS023	Environment	Cycle schedule	Day, season, tide or custom phase cycle with explicit clock domain.
GNS024	Environment	Habitat support	Where an organism, trait or action may operate under the selected profile.

ID	Family	Mechanism	Core contract
GNS025	Operator	Expression	Genome + stage + environment → trait/development proposal with trace.
GNS026	Operator	Growth	Consumes resources and proposes body/state changes under the development graph.
GNS027	Operator	Metabolism	Typed resource conversion with conservation/budget and capability contract.
GNS028	Operator	Sense	Environment/other-state query producing an uncertain organism-local observation.
GNS029	Operator	Act	Movement, feeding, signaling, defense or construction proposal under support and cost guards.
GNS030	Operator	Reproduce	Asexual, sexual, budding or custom offspring proposal with parent and resource contracts.
GNS031	Operator	Mutate	Seeded/profile-bound genome edit proposal with scope, rate and reason trace.
GNS032	Operator	Recombine	Profile-bound parent genome mapping with explicit ordering and crossover semantics.
GNS033	Relation	Parent-of	Birth-lineage edge; adoption, ownership or social role are separate relations.
GNS034	Relation	Consumes	Organism/action to resource relation with quantity and time.
GNS035	Relation	Predates	Profile-specific feeding relation; does not imply every encounter produces a kill.
GNS036	Relation	Competes-with	Shared limiting-resource relation with a declared competition model.
GNS037	Relation	Mutualism	Reciprocal benefit relation under declared exchanges and conditions.
GNS038	Invariant	Nonnegative inventory	No commit may consume more authoritative resource/matter than available.
GNS039	Invariant	Conservation profile	Selected matter/energy/resource balances and external-source terms are explicit.
GNS040	Guard	Viability	Development/body/state satisfies required finite constraints for the selected life profile.
GNS041	Guard	Reproduction readiness	Stage, resources, compatibility, timing, capacity and authorization pass.
GNS042	Guard	Ecological interaction	Support, participant state, relation profile and conflict ordering pass.
GNS043	Event	Birth	Atomic offspring creation, parent/resource patch, stable ID and ancestry lineage.
GNS044	Event	Death/decomposition	Atomic life-state terminal transition plus released-resource and lineage effects.
GNS045	Certificate	Organism transition certificate	Dependencies, budgets, guards, random draws, pre/post hashes and capability.
GNS046	Packing	Genome/population codebook	Local hot indices bound to cold atlas/profile hashes; not complete biological identity.
GNS047	Query	Ancestry and ecology query	Ancestors, descendants, trait provenance, resource flows, niches and first divergence.
GNS048	Adapter	External biology/ecology oracle	Versioned scientific, simulation or educational library behind typed proposal-only ports.

Claims Ledger

Claim	Disposition	Reason
GENESIS can encode a finite synthetic genome, organism state and ancestry graph.	ADMIT	The records and transitions are explicitly typed and bounded.
A fixed seed and PRNG profile reproduce the same stochastic fixture.	ADMIT	Only for the same algorithms, ordering, budgets and external inputs.
A seed reconstructs unstored measurements, interventions or living systems.	REJECT	External novelty and evidence must remain stored.
Genome identity equals organism identity.	REJECT	Many organisms may share a genome; life history and state remain distinct.
Phenotype is determined by genome alone.	REJECT	Development, stage, environment and stochastic/profile inputs are explicit.
The cartridge proves a universal theory of evolution.	REJECT	It supplies profile-bound simulation and lineage mechanics only.
Selection always optimizes toward a best organism.	REJECT	Fitness is contextual, multi-objective and profile-dependent; drift and extinction may occur.
Conservation and budget checks can certify selected synthetic transitions.	ADMIT	Within the declared quantities, external source terms and arithmetic profile.
GENESIS can replace validated population genetics or ecological software.	REJECT	Mature tools belong behind versioned adapters.
Mutation is random in an absolute physical sense.	REJECT	The runtime uses declared stochastic sources; randomness claims remain scoped.
The cartridge is suitable for medical, reproductive or gene-editing decisions.	REJECT	No clinical, biological-safety or regulatory validation is claimed.
A generated creature image proves the underlying organism record.	REJECT	Rendering is a downstream projection and may omit hidden state.

High-Impact Application Profiles

Profile	What the cartridge enables
Artificial-life sandbox	Build evolving creature worlds whose bodies, resources, behavior and ancestry remain inspectable.
Creature constructor / monster breeder	Make modular body plans and inheritance into playable mechanics rather than opaque stat rolls.
Procedural ecosystem game	Run food webs, seasons, extinctions, migrations and player interventions with replayable causes.
Evolution puzzle laboratory	Ask players to satisfy environmental constraints through bounded mutation and development operators.
Educational model bridge	Connect simplified classroom worlds to external scientific tools without blurring the evidence boundary.

Final Formal Definition

FORMAL DEFINITION

UGTOMS GENESIS 1.0 is a component-scoped synthetic-life cartridge in which world profiles, genomes, loci, alleles, regulatory dependencies, development modules, traits, body graphs, life stages, topology changes, energy and matter ledgers, sensors, actions, reproduction, mutation, recombination, organisms, populations, niches, ecological relations, stochastic schedules, candidate variations, verified births/deaths/interactions, checkpoints and ancestry lineage are typed substrate data; failed life proposals remain non-authoritative evidence; and only verified atomic events may extend a living world.

Implementation sequence

1. Freeze Genome, DevelopmentModule, OrganismState, PopulationState, EnvironmentField, LifeEvent and TransitionCertificate schemas.
2. Implement the Glassreef fixture with exact resource accounting, rejected-candidate isolation and ancestry replay.
3. Add RULEFORGE gameplay adapters and a compact browser viewer for genomes, bodies and lineages.
4. Add optional REACT resource-conversion and CIVITAS habitat adapters through explicit capability maps.
5. Admit external scientific models only with versioned units, uncertainty, validation scope and proposal-only authority.

RELEASE CONCLUSION

This 1.0 document freezes a domain-facing contract. It is deliberately useful before a production engine exists: implementations can now disagree visibly through typed profiles, reason codes and certificates instead of hiding differences behind the same symbol.